

Local tumour control and eye preservation after gamma-knife radiosurgery of choroidal melanomas

Werner Wackernagel,^{1,2} Etienne Holl,^{2,3} Lisa Tarmann,^{1,2} Christoph Mayer,^{1,2} Alexander Avian,⁴ Mona Schneider,^{1,2} Karin S Kapp,^{2,5} Gerald Langmann^{1,2}

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¹Department of Ophthalmology, Medical University Graz, Graz, Austria
²Comprehensive Cancer Center Graz, Graz, Austria
³Department of Neurosurgery, Medical University Graz, Graz, Austria
⁴Institute for Medical Informatics, Statistics and Documentation, Medical University Graz, Graz, Austria
⁵Department of Therapeutic Radiology and Oncology, Medical University Graz, Graz, Austria

Correspondence to
 Dr Werner Wackernagel, Department of Ophthalmology, Medical University Graz, Auenbruggerplatz 4, A-8036 Graz, Austria; werner.wackernagel@medunigraz.at

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ABSTRACT

Background/aims To report on local tumour control and eye preservation after gamma knife radiosurgery (GK-RS) to treat choroidal melanomas.

Methods A total of 189 patients with choroidal melanoma were treated with GK-RS, with treatment doses between 25 and 80 Grays. The main outcome measures of our retrospective analysis were local tumour control, time to recurrence, eye retention rate and the reason for and time to secondary enucleation. Patient-associated, tumour-associated and treatment-associated parameters were evaluated as potential risk factors.

Results Local tumour control was achieved in 94.4% of patients. The estimated tumour control rates were 97.6% at 1 year, 94.2% at 5 years and 92.4% at 10 years after treatment. Recurrence was observed between 3.1 months and 60.7 months post-treatment (median: 13.5 months). Advanced tumour stage (Tumour, Node, Metastasis (TNM) 3–4) was the most important risk factor for recurrence (Fine-Gray model; subhazard ratio, SHR: 3.3; $p=0.079$). The treatment dose was not related to tumour recurrence. The eye preservation rate was 81.6% at 5 years after treatment, remaining stable thereafter. Twenty-five eyes (14.1%) had to be enucleated at between 17 days and 68.0 months (median: 13.9 months) after GK-RS, and advanced tumour stage (Cox model; $p=0.005$), treatment dose ($p=0.048$), pretreatment visual acuity ($p=0.016$), and retinal detachment ($p=0.027$) were risk factors for requiring enucleation.

Conclusions GK-RS achieved a high tumour control rate, comparable to linear accelerator-based radiotherapy. Advanced TNM stage was a predictive risk factor for tumour recurrence and for secondary enucleation after GK-RS. Lower treatment doses were unrelated to tumour recurrence, although they were associated with an improved eye retention rate.

INTRODUCTION

The treatment of melanoma has changed significantly over past decades. Although enucleation was the treatment of choice for most medium and large melanomas until the 1980s, brachytherapy (using iodine-125 or ruthenium-106) and proton beam radiotherapy are the most commonly used treatment modalities today.¹ The main goals of treatment are ensuring local tumour control and preservation of the eye. The Collaborative Ocular Melanoma Study failed to show a difference in survival after enucleation versus iodine-125 brachytherapy for medium-sized melanomas and paved the way for globe-preserving treatments.² Mortality appears to depend primarily on intrinsic tumour characteristics, such as cytogenetic status, and less on treatment modality.³

Some smaller studies, however, found an association between increased mortality due to systemic metastasis and tumour recurrence after globe preserving treatment.^{4–5} Chang and McCannel recently published a review of the literature on local treatment failure after different treatment modalities and emphasised the importance of local tumour control.⁶ Very high control rates above 95% have been reported for proton beam radiotherapy and iodine-125 plaque brachytherapy, with excellent eye retention rates.^{5–7–8} Reports on outcomes after photon-ray and gamma-ray based radiotherapy are sparse in comparison and include only a small number of patients.^{9–12} In our study, we report long-term data on local tumour control and eye preservation in patients with posterior uveal melanoma treated with single-fraction gamma-knife radiosurgery (GK-RS).

PATIENTS AND METHODS

Study design

Our study comprised 189 patients with choroidal melanoma who were treated with GK-RS at the Medical University of Graz between June 1992 and May 2010. All patients were offered GK-RS as an alternative to enucleation if they wished to retain their eye, and other globe-preserving treatment options were not feasible because of tumour size, tumour location or the patient's general health. Proton beam radiotherapy was not available in Austria at that time. All patients were fully informed about the treatment procedure and provided informed consent before treatment. Institutional review board approval was obtained to retrospectively analyse radiosurgery treatment plans and medical records. Data had been recorded prospectively in electronic databases at the Department of Neurosurgery and at the Department of Ophthalmology. The patients were excluded from the analysis if the melanoma had previously been treated with other treatment modalities (transpupillary thermotherapy, 3 patients; ruthenium plaque brachytherapy, 3 patients) or if clinical data were incomplete (6 patients). Details on the patient and tumour characteristics of the 177 patients remaining in the study have been published previously and are summarised in table 1.¹³

Preirradiation assessment

At the initial clinical evaluation, all patients underwent a full ophthalmic examination, including medical and family histories, best-corrected Snellen visual acuity, slit-lamp and fundus examinations, measurement of intraocular pressure, wide-angle fundus photography and standardised echography

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Table 1 Treatment and tumour parameters

Patient parameters	Value
Sex	
Male	91 (51.41%)
Female	86 (48.59%)
Laterality	
Right	83 (46.89%)
Left	94 (52.11%)
Age (years)	62.1 (median; IQR: 51.8–70.5)
Diabetes	22 (12.4%)
Systemic hypertension	62 (35.0%)
Coronary artery disease	25 (14.1%)
Other cancer	12 (6.8%)
Tumour parameters	Value
Tumour height (mm)	6.1 (median; IQR: 4.4–8.2)
Longest basal diameter (mm)	11.7 (median; IQR: 9.4–14.3)
TNM 7 classification	
T1	28 (15.8%)
T2	74 (41.8%)
T3	71 (40.1%)
T4	4 (2.3%)
Distance to optic disc margin (mm)	3.0 (median; IQR: 0.5–4.5)
Distance to foveola (mm)	1.5 (median; IQR: 0–3.5)
Shape	
Dome	135 (76.3%)
Mushroom	42 (23.7%)
Retinal detachment	
Present	108 (61.0%)
Absent	69 (39.0%)
Tumour pigmentation	
Melanotic	165 (93.2%)
Amelanotic	12 (6.8%)
Coronal location	
Nasal	28 (15.8%)
Midline	59 (33.33%)
Temporal	90 (50.85%)
Sagittal location	
Superior	49 (27.7%)
Horizontal	80 (45.2%)
Inferior	48 (27.1%)
Anterior margin	
Ciliary body	7 (4.0%)
Anterior to equator	49 (27.7%)
Posterior to equator	121 (68.4%)
Treatment parameters	Value
Treatment dose (Gy)	30 (Median, range: 25–80)
Isodose	50 (Median, range: 32–80)
High dose (≥ 35 Gy):	N=66 (median dose: 45 Gy, range: 35–80 Gy)
Low dose: (< 35 Gy):	N=111 (median dose: 30 Gy, range: 25–30 Gy)

(A- and B-scans). Angiography (fluorescein and indocyanine green) and optical coherence tomography were performed if necessary. Tumours were classified according to the Tumour, Node, Metastasis (TNM) 7 classification and divided into two groups (TNM1–2 and TNM 3–4).¹⁴ Systemic examinations (physical examination, blood work including liver parameters, chest X-ray, abdominal ultrasonography, and CT scans and/or MRI of the liver) were performed before treatment to rule out distant metastases.

Treatment

GK-RS was performed as described previously.¹³ For immobilisation of the eye, a retrobulbar anaesthetic injection was given, and the four rectus muscles were fixed to a stereotactic frame with transconjunctival sutures. MRI was performed using axial and coronal sections (T1 using a contrast-enhancing agent, T2 if needed). Leksell GammaPlan Software (Elekta, Stockholm, Sweden) was used for treatment planning. The visible tumour volume was identified on MRI and drawn slice-by-slice (gross tumour volume, GTV). A safety margin of 1 mm was added to create the planning target volume (PTV). The median treatment dose was 30 Gy (range: 25–80 Gy; table 1). Critical intraocular structures were spared if possible, without compromising coverage of the PTV. Multiple collimators of different sizes and collimator plugs were used to minimise radiation to the contralateral eye and provide conformal coverage of the PTV.

Follow-up

The patients were re-examined approximately 1 month after treatment. Thereafter, they were followed every 3 months for the first 2 years, every 6 months up to 5 years after treatment and then annually. At each follow-up, the ophthalmic examination was repeated. Systemic re-examinations to detect metastatic disease were performed every 6 months (blood work including liver function tests and liver ultrasonography) or annually (chest X-ray; abdominal MRI/CT).

Outcome measures

The primary outcome measures of the study were local tumour control and secondary enucleation. Tumour recurrence was defined as (1) an increase in tumour height on ultrasound at least 6 months after irradiation that was confirmed at two subsequent examinations or (2) an increase of the tumour basal diameter after treatment in comparison with previous photographs at any time after treatment. For analysis of eye preservation, the reason for and time to secondary enucleation were evaluated.

Statistics

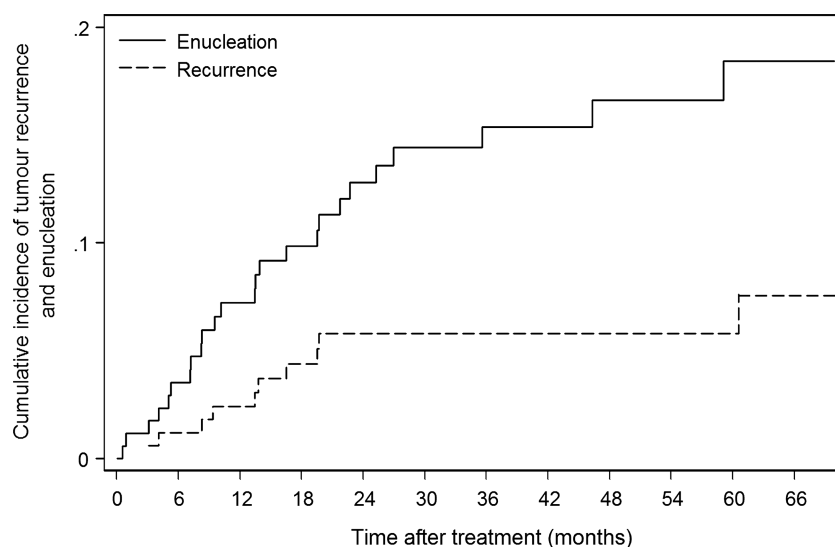
The values of continuous variables are presented as the median, range and (IQR). Categorical variables are described as absolute and relative frequencies. Follow-up time was described as the median and IQR. The occurrence of tumour recurrence over time in the presence of enucleation as a competing risk is presented as a cumulative incidence function (CIF). Comparison of CIFs between groups was performed using Gray's test (R 2.14.2 using the cmprsk package, V2.2-3 and the etm package, V0.5-4, and Stata V12.1). Risk factors for tumour recurrence were analysed using a competing risk regression (Fine-Gray model). Kaplan–Meier estimates were calculated for the risk of secondary enucleation (reported as 1-survival to be comparable to CIFs calculated for recurrence), and a Cox regression model was used to determine risk factors and estimate hazard ratios. The χ^2 test and Fisher's exact test were used to compare categorical variables between groups.

RESULTS

Ocular status before treatment

Before treatment 58, patients (32.8%) had a visual acuity of 20/40 or better on the affected eye, and 128 patients (72.3%) had a visual acuity of 20/200 or better. The visual acuity of the fellow eye was equal to or better than 20/40 in 165 cases (93.2%). On the affected eye, 108 patients (61.0%) showed retinal detachment, 41 (23.2%) showed cataract, 6 (3.4%) were

Figure 1 Incidence of tumour recurrence (cumulative incidence) and enucleation (1-Survival).



pseudophakic, 6 (3.4%) had diagnosed glaucoma, 7 (4.0%) showed age-related macular degeneration and 4 (2.3%) showed diabetic retinopathy.

Treatment characteristics

The median treatment dose was 30 Gy (range: 25–80 Gy). Sixty-six patients (37.3%), all treated before 7 May 2003, received doses of 35–80 Gy (high-dose group). Thereafter, 30 Gy (87 patients, 49.2%) and 25 Gy (24 patients, 13.6%) were used as treatment doses (low-dose group). Treatment was planned for the 50% isodose (median: 50%; range: 32–80%; table 1).

Follow-up

The median overall follow-up was 39.5 months (IQR: 20.6–70.5 months). The median follow-up of the high-dose group was 62.3 (IQR: 24.5–121.6) months, whereas the follow-up was 34.9 (IQR: 19.2–55.3) months in the low-dose group.

Recurrence

During follow-up, tumour control was achieved in 167 patients (94.4%). Ten patients (5.6%) experienced tumour recurrence between 3.1 months and 60.7 months post-treatment. In two cases, recurrence was observed at the tumour margin only, whereas it presented as an increase in height or general tumour growth in eight cases. The cumulative incidences of recurrence (accounting for the competing risk of enucleation for other causes) after 1, 2, 5 and 10 years were 2.4% (95% CI 0.8% to 5.7%), 5.8% (95% CI 2.8% to 10.3%), 5.8% (95% CI 2.8% to 10.3%) and 7.6% (95% CI 3.6% to 13.6%), respectively (figure 1).

Five eyes were enucleated due to recurrence without attempting additional salvage treatment (table 2). Four eyes were treated with additional conservative treatment, and two of the four could be salvaged (one by additional GK-RS, one by transpupillary thermotherapy). Two eyes were enucleated after additional conservative treatment because of treatment-related side effects: one after Ru106 plaque brachytherapy and one after re-treatment with GK-RS and additional transpupillary thermotherapy. One patient with local recurrence received no further treatment because of advanced systemic metastatic disease. Competing risk regression analysis (univariate Fine–Gray model) showed a trend for an increased risk for recurrence only for advanced tumour stage (TNM 3–4 vs TNM1–2; $p=0.079$,

sub-HR, SHR: 3.3). Treatment dose ($p=0.572$, SHR: 1.4), other tumour parameters (location, shape, distance to disc or fovea) and patient parameters (age, sex, concurrent systemic diseases, pretreatment visual acuity) were not significant risk factors for tumour recurrence (figure 2).

Secondary enucleation

In total, 25 eyes (14.1%) were enucleated between 17 days and 68.0 months after GK-RS (median: 13.9 months). Seven eyes were enucleated after tumour recurrence, and 18 eyes were enucleated for radiation-induced complications (without recurrence). The risk for enucleation was 7.2% (95% CI 4.2% to 12.4%) after 1 year, 12.8% (95% CI 8.4% to 19.2%) after 2 years and 18.4% (95% CI 12.5% to 26.8%) after both 5 and 10 years (figure 1). After tumour recurrence, the most frequent single cause for enucleation was uncontrollable neovascular glaucoma (12 cases). However, most patients with secondary enucleation encountered multiple radiation-induced complications, such as optic neuropathy (8 cases), vitreous haemorrhage (7 cases), persistent retinal detachment (17 cases), radiation maculopathy (11 cases) and retinopathy (14 cases), requiring one or more adjunct treatments in the majority of patients ($n=114$, 64.4%; table 2) and leading to enucleation in another six cases.

Risk factors for enucleation identified by univariate cox regression analysis were tumour recurrence ($p<0.001$; HR: 11.7), advanced tumour stage (TNM3–4 vs TNM 1–2; $p=0.005$; HR:

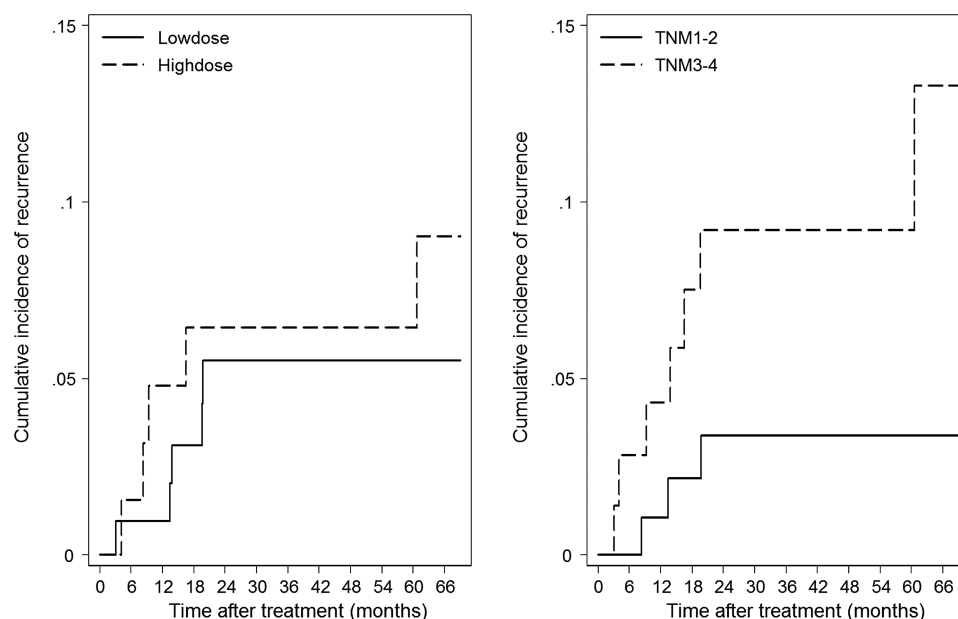
Table 2 Adjunct treatments for tumour recurrence

Treatment modality	Number (%) months after initial treatment	Final outcome salvage/enucleation (months after initial treatment)
Brachytherapy (Ru 106)*	1 (0.6) 8.4	–/1 (19.0)
Gamma-Knife Radiosurgery	2 (1.1) 63.2 65.3	1/1 (68.0)
TTT	2 (1.1) 13.8 20.7	1/1 (68.0†)

*Brachytherapy was performed in one additional case immediately after incomplete gamma knife radiosurgery (because of an inability of the patient to keep the head in a reclined position) without evidence of tumour recurrence.

†The eye was enucleated (68.0 months after initial treatment) after adjunct transpupillary thermotherapy (TTT) (at 20.7 months) and adjunct GK-RS (at 65.3 months).

Figure 2 Cumulative incidence function for tumour recurrence (accounting for enucleation as a competing risk) according to (A) treatment dose and (B) TNM stage.



3.3), decreased pretreatment visual acuity ($p=0.016$; HR per Snellen line: 1.12) and retinal detachment before treatment ($p=0.027$, HR: 3.0). Tumour shape (mushroom-like appearance) was significant ($p<0.001$, HR: 4.1), but it was correlated with advanced tumour stage (TNM3-4) and therefore excluded. The treatment dose (low-dose group, 25–30 Gy vs high-dose group, 35–80 Gy) was marginally significant ($p=0.048$, HR: 2.3; figure 3). The rates of tumour recurrence and secondary enucleation according to treatment dose and TNM tumour stage are summarised in table 3.

DISCUSSION

Stereotactic radiotherapy (SRT) and GK-RS have been used for the treatment of uveal melanoma for approximately two decades, primarily in centres with no access to proton beam radiotherapy.^{9–12 15 16} We have been using GK-RS as an alternative to enucleation when plaque brachytherapy was not

indicated due to large tumour size, location of the melanoma close to the optic nerve, or the patient's general health. GK-RS delivers high doses of radiation to the tumour, with a steep dose fall-off, thus sparing surrounding healthy tissue. However, there is significant dose inhomogeneity within the target volume, as the treatment is typically planned to the 50% isodose. Initially, very high treatment doses were chosen to ensure complete tumour destruction, as little was known about the radiobiological effectiveness of GK-RS treatment on uveal melanoma. Over time, we lowered the treatment dose in a stepwise manner to reduce radiation-induced side effects.¹⁷ This is in accordance with reports by other authors, who reduced the treatment dose and successfully applied low-dose regimens for stereotactic radiosurgery.^{11 18 19} Uveal melanoma is generally considered a radioresistant tumour that responds well to high-dose hypofractionated treatment.²⁰ In our series, radiosurgery achieved a high rate of local tumour control, which was notably unrelated to

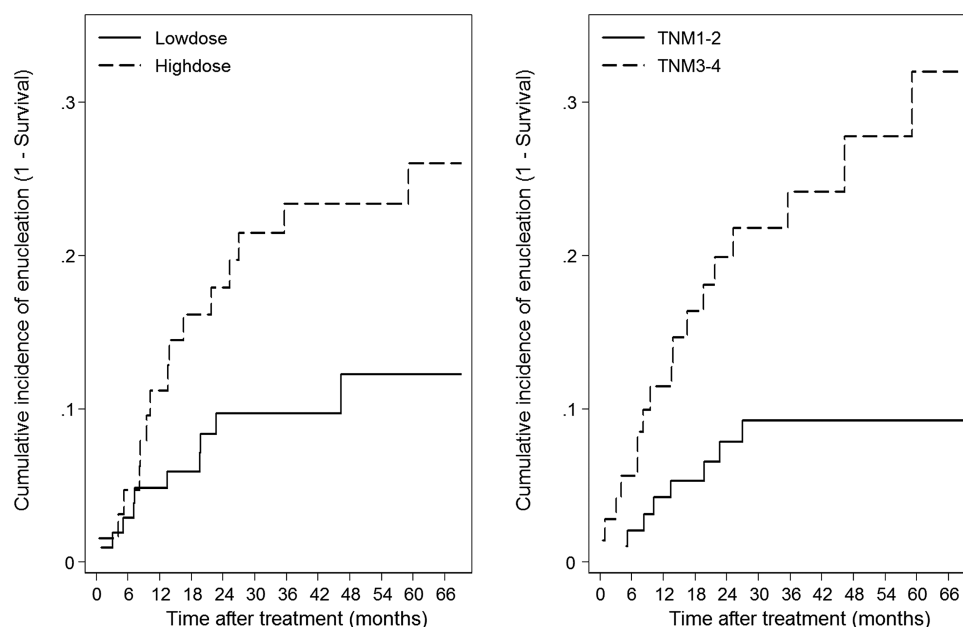


Figure 3 Risk for enucleation (1-Survival) according to (A) treatment dose and (B) TNM category.

Table 3 Tumour recurrence and secondary enucleation, according to treatment dose and tumour stage

		1 year (%) (95% CI)	2 years (%) (95% CI)	5 years (%) (95% CI)	10 years (%) (95% CI)
Recurrence					
p=0.572	Low dose	3.1 (0.8 to 8.0)	5.5 (0.2 to 11.5)	5.5 (0.2 to 11.5)	5.5 (0.2 to 11.5)
	High dose	4.8 (1.3 to 12.0)	6.4 (2.1 to 14.3)	6.4 (2.1 to 14.3)	9.0 (3.2 to 18.6)
p=0.079	TNM1–2	1.1 (0.1 to 5.2)	3.4 (0.9 to 8.8)	3.4 (0.9 to 8.8)	3.4 (0.9 to 8.8)
	TNM3–4	4.2 (1.1 to 10.9)	9.2 (3.7 to 17.7)	9.2 (3.7 to 17.7)	13.3 (5.1 to 25.3)
	Total	2.4 (0.8 to 5.7)	5.8 (2.8 to 10.3)	5.8 (2.8 to 10.3)	7.6 (3.6 to 13.6)
Enucleation					
p=0.048	Low dose	4.8 (2.0 to 11.1)	9.6 (5.1 to 17.8)	12.2 (6.4 to 22.6)	12.2 (6.4 to 22.6)
	High dose	11.1 (5.5 to 22.0)	17.9 (10.3 to 30.0)	26.0 (16.4 to 39.7)	26.0 (16.4 to 39.7)
p=0.005	TNM1–2	4.2 (1.6 to 10.8)	7.8 (3.8 to 15.8)	9.2 (4.7 to 17.7)	9.2 (4.7 to 17.7)
	TNM3–4	11.4 (5.8 to 21.5)	19.8 (11.9 to 31.8)	31.9 (20.0 to 48.5)	31.9 (20.0 to 48.5)
	Total	7.2 (4.2 to 12.4)	12.8 (8.4 to 19.2)	18.4 (12.5 to 26.8)	18.4 (12.5 to 26.8)

p Values (univariate Fine–Gray model) for intergroup comparisons (low vs high dose; TNM 1–2 vs TNM 3–4).

treatment dose and patient parameters but dependent upon TNM 7 tumour stage. The eye preservation rate was less favourable, and secondary enucleation was required in 25 cases, primarily because of radiation-related side effects. In contrast to tumour recurrence, secondary enucleation was dependent on both treatment dose and TNM tumour stage. The strengths of our study are the high number of included patients and the long follow-up. However, as tumour recurrence and enucleation occurred in only a few patients, some risk factors might have gone undetected. This and the influence of adjunct treatments on the risk of enucleation are possible weaknesses of our study.

Local tumour recurrence has been associated with an increased risk of metastasis. Whether local tumour recurrence is related to the development of metastatic disease or simply more likely to occur in more aggressive tumours remains unclear.⁶ The reason for tumour recurrence is not easily discernible and only a few authors have addressed this issue.⁵ Parts of the tumour may be missed by radiotherapy and continue to grow after treatment, usually at one tumour margin (figure 4). Alternatively, the tumour may be adequately encompassed by the radiation field, with some radioresistant cells surviving, resulting in a temporary growth arrest and infield recurrence later on. As those cells might acquire additional genetic aberrations over time, the reason for and time to tumour recurrence should perhaps be considered when analysing the impact of tumour recurrence on mortality. For the analysis of eye preservation, we first considered an approach similar to Foss *et al*²¹, excluding cases with enucleation after tumour recurrence from the Cox/Fine–Gray model. However, tumour recurrence was the second most frequent cause of enucleation in our series, and there were two cases of enucleation for treatment-related complications after salvage treatment for tumour recurrence. We therefore decided to follow the methodology of Egan *et al*²² and Dunavoelgyi *et al*⁹ and included all cases of enucleation in the analysis.

Our findings compare well to previous reports on external radiotherapy and GK-RS by other authors. Modorati *et al*¹¹ and Sarici and Pazarli²³ found local tumour control rates after GK-RS of 91.0% and 90.0% and eye retention rates of 89.7% and 82% after median follow-ups of 31.3 and 40 months, respectively. Recently, Dunavoelgyi *et al*⁹ published results after linear accelerator-based hypofractionated SRT. Their reported rates of local tumour control (95.9% at 5 years after treatment) and eye preservation (78.6% at 5 years) are similar to our results. However, the authors failed to show an association

between treatment dose and the risk for tumour recurrence. They also did not find a relation between dose and time to enucleation, which was observed in our study.⁹ Krema *et al*¹⁰ also

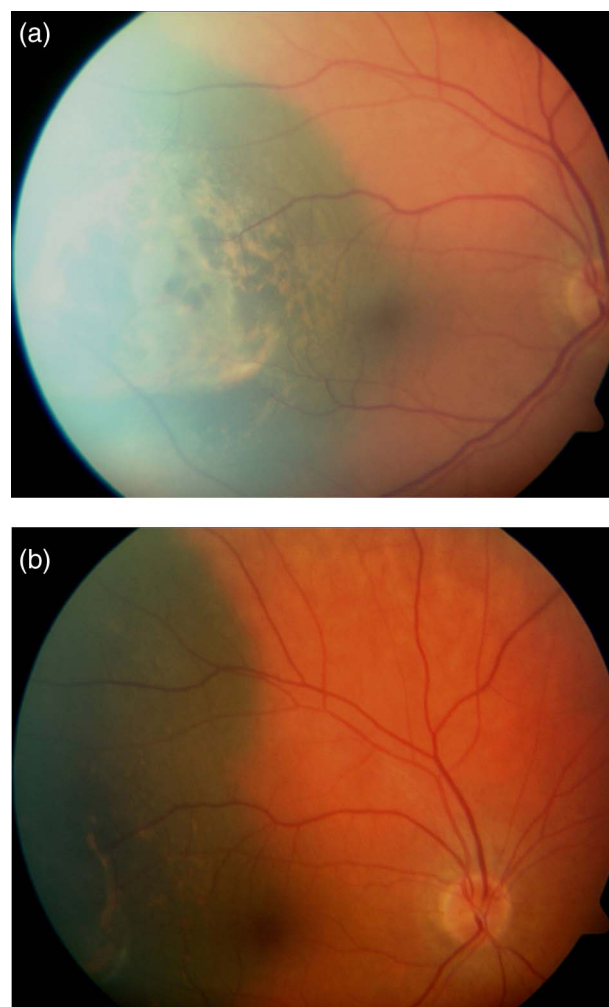


Figure 4 (A) Choroidal melanoma of the right eye involving the macula before treatment (25 Gy, 50% isodose). (B) Local tumour recurrence at the superior tumour margin 12 months after gamma knife radiosurgery, which was successfully treated with additional transpupillary thermotherapy, and the tumour remained stable thereafter. The patient developed metastatic disease 5.4 years after treatment. Access the article online to view this figure in colour.

reported a high tumour control rate of 94% at 37 months after hypofractionated radiotherapy for juxtapapillary melanomas in their retrospective study. The only prospective study on hypofractionated SRT to date was published by Muller *et al.*¹² Their local tumour control rate of 96% after a median follow-up period of 32 months confirms that excellent local tumour control can be achieved with SRT and is comparable with the results obtained after proton beam radiotherapy.^{12 24 25} However, the rates of secondary enucleation reported by Muller *et al.* (15 of 102 eyes) and all other reports after SRT are worse when compared with the eye preservation rates achieved after proton beam radiotherapy.⁸ Unfortunately, the patient selection criteria, tumour size and location, and the application of adjunct treatments differed vastly between these reports, and the results are therefore not comparable between studies. In addition, fractionation schemes for external beam radiotherapy also differ, making direct comparisons with GK-RS (single fraction) extremely difficult. Chang and McCannel have recently published a review of local tumour control after different treatment modalities.⁶ They found large variability among outcomes after different treatment methods and even between the outcomes reported for the same treatment method, attributable—at least in part—to the heterogeneity of the studies regarding inclusion criteria.

Despite the limitations in comparability, our results suggest that GK-RS is an alternative treatment option to avoid enucleation, providing a viable chance of local tumour control and eye preservation in cases considered unsuitable for plaque brachytherapy when proton beam radiotherapy is unavailable. The TNM category was identified as the most important risk factor for secondary enucleation for tumour recurrence and should thus be considered a predictive parameter. Our results further support previous reports, which failed to show an impact of lower treatment dose on tumour control but did observe improved eye preservation. Thus, our findings encourage further efforts to identify the optimal treatment dose for the radiotherapy of choroidal melanoma.^{17 18}

Contributors All authors substantially contributed to preparation of the manuscript (according to ICMJE requirements). In detail, their authorship is based on contribution to: WW: conception and design of the study, data acquisition, analysis and interpretation of data and drafting and revising the manuscript. EH, LT, CM and MS: data acquisition and drafting and revising the manuscript. AA: conception and design of the study, analysis and interpretation of data and revising the manuscript. KSK and GL: conception and design of the study, interpreting the results and revising the manuscript.

Competing interests None.

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REFERENCES

- Singh AD, Turell ME, Topham AK. Uveal melanoma: trends in incidence, treatment, and survival. *Ophthalmology* 2011;118:1881–5.
- Singh AD, Kivela T. The collaborative ocular melanoma study. *Ophthalmol Clin North Am* 2005;18:129–42, ix.
- Prescher G, Bornfeld N, Hirsch H, *et al.* Prognostic implications of monosomy 3 in uveal melanoma. *Lancet* 1996;347:1222–5.
- Bechrakis NE, Petousis V, Willerding G, *et al.* Ten-year results of transscleral resection of large uveal melanomas: local tumour control and metastatic rate. *Br J Ophthalmol* 2010;94:460–6.
- Gragoudas E, Li W, Goitein M, *et al.* Evidence-based estimates of outcome in patients irradiated for intraocular melanoma. *Arch Ophthalmol* 2002;120:1665–71.
- Chang MY, McCannel TA. Local treatment failure after globe-conserving therapy for choroidal melanoma. *Br J Ophthalmol* 2013;97:804–11.
- Damato B, Kacperek A, Chopra M, *et al.* Proton beam radiotherapy of choroidal melanoma: the Liverpool-Clatterbridge experience. *Int J Radiat Oncol Biol Phys* 2005;62:1405–11.
- Egger E, Zografos L, Schalenbourg A, *et al.* Eye retention after proton beam radiotherapy for uveal melanoma. *Int J Radiat Oncol Biol Phys* 2003;55:867–80.
- Dunavoelgyi R, Dieckmann K, Gleiss A, *et al.* Local tumor control, visual acuity, and survival after hypofractionated stereotactic photon radiotherapy of choroidal melanoma in 212 patients treated between 1997 and 2007. *Int J Radiat Oncol Biol Phys* 2011;81:199–205.
- Krema H, Somani S, Sahgal A, *et al.* Stereotactic radiotherapy for treatment of juxtapapillary choroidal melanoma: 3-year follow-up. *Br J Ophthalmol* 2009;93:1172–6.
- Modorati G, Miserocchi E, Galli L, *et al.* Gamma knife radiosurgery for uveal melanoma: 12 years of experience. *Br J Ophthalmol* 2009;93:40–4.
- Muller K, Naus N, Nowak PJ, *et al.* Fractionated stereotactic radiotherapy for uveal melanoma, late clinical results. *Radiother Oncol* 2012;102:219–24.
- Wackernagel W, Holl E, Tarmann L, *et al.* Visual acuity after Gamma-Knife radiosurgery of choroidal melanomas. *Br J Ophthalmol* 2013;97:153–8.
- Kivela T, Kujala E. Prognostication in eye cancer: the latest tumor, node, metastasis classification and beyond. *Eye* 2013;27:243–52.
- Emara K, Weisbrod DJ, Sahgal A, *et al.* Stereotactic radiotherapy in the treatment of juxtapapillary choroidal melanoma: preliminary results. *Int J Radiat Oncol Biol Phys* 2004;59:94–100.
- Dinca EB, Yianni J, Rowe J, *et al.* Survival and complications following gamma knife radiosurgery or enucleation for ocular melanoma: a 20-year experience. *Acta Neurochirurgica* 2012;154:605–10.
- Langmann G, Pendl G, Mullner K, *et al.* High-compared with low-dose radiosurgery for uveal melanomas. *J Neurosurg* 2002;97:640–3.
- Schirmer CM, Chan M, Mignano J, *et al.* Dose de-escalation with gamma knife radiosurgery in the treatment of choroidal melanoma. *Int J Radiat Oncol Biol Phys* 2009;75:170–6.
- Mueller AJ, Talies S, Schaller UC, *et al.* Stereotactic radiosurgery of large uveal melanomas with the gamma-knife. *Ophthalmology* 2000;107:1381–7; discussion 7–8.
- van den Aardweg GJ, Kilic E, de Klein A, *et al.* Dose fractionation effects in primary and metastatic human uveal melanoma cell lines. *Invest Ophthalmol Vis Sci* 2003;44:4660–4.
- Foss AJ, Whelehan I, Hungerford JL, *et al.* Predictive factors for the development of rubeosis following proton beam radiotherapy for uveal melanoma. *Br J Ophthalmol* 1997;81:748–54.
- Egan KM, Gragoudas ES, Seddon JM, *et al.* The risk of enucleation after proton beam irradiation of uveal melanoma. *Ophthalmology* 1989;96:1377–82; discussion 82–3.
- Sarici AM, Pazarli H. Gamma-knife-based stereotactic radiosurgery for medium- and large-sized posterior uveal melanoma. *Graefes Arch Clin Exp Ophthalmol* 2013;251:285–94.
- Gragoudas ES, Lane AM, Munzenrider J, *et al.* Long-term risk of local failure after proton therapy for choroidal/ciliary body melanoma. *Trans Am Ophthalmol Soc* 2002;100:43–8; discussion 8–9.
- Caujolle JP, Mammar H, Chamorey E, *et al.* Proton beam radiotherapy for uveal melanomas at nice teaching hospital: 16 years' experience. *Int J Radiat Oncol Biol Phys* 2010;78:98–103.